Lakshya JEE 2.0 2025

Maths

Inverse Trigonometric Functions

DPP: 5

Q1 Prove that
$\cos^{-1}\frac{4}{5}+\cos^{-1}\frac{12}{13}=\cos^{-1}\frac{33}{65}$

Q2 Find x, if
$\tan^{-1}(2+x)+\tan^{-1}(2-x)=\tan^{-1}\frac{2}{3}$

Q3 If $\sin^{-1}(1-x)-2\sin^{-1}x=\frac{\pi}{2}$, then x is equal to
(A) $0,-\frac{1}{2}$
(B) $0,\frac{1}{2}$
(C) 0
(D) none of these

Q4 If
$\tan^{-1}(x-1)+\tan^{-1}x+\tan^{-1}(x+1)=\tan^{-1}3x$ then x is equal to

(A) $\pm\frac{1}{2}$
(B) $0,\frac{1}{2}$
(C) $0,-\frac{1}{2}$
(D) $0,\pm\frac{1}{2}$

Q5 If $\sin^{-1}\left(\frac{2a}{1+a^2}\right)-\cos^{-1}\left(\frac{1-b^2}{1+b^2}\right)=\tan^{-1}\left(\frac{2x}{1-x^2}\right)$
then the value of x is
(A) a
(B) b
(C) $\frac{a+b}{1-ab}$
(D) $\frac{a-b}{1+ab}$

Q6 If $\tan^{-1}2$ and $\tan^{-1}3$ are two angles of a triangle, then the third angle is
(A) $\frac{\pi}{2}$
(B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$
(D) $\frac{\pi}{6}$

Q7 If x, y, z are in A.P. and $\tan^{-1}x, \tan^{-1}y$ and $\tan^{-1}z$ are also in A.P., then
(A) $x=y=z$
(B) $2x=3y=6z$
(C) $6x=3y=2z$
(D) $6x=4y=3z$

Q8 If S is the sum of the first 10 terms of the series $\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+\tan^{-1}\left(\frac{1}{21}\right)+\ldots\ldots$ then $\tan(S)$ is equal to
(A) $\frac{10}{11}$
(B) $\frac{5}{11}$
(C) $-\frac{6}{5}$
(D) $\frac{5}{6}$

Q9 A possible value of $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)$ is
(A) $\sqrt{7}-1$
(B) $\frac{1}{\sqrt{7}}$
(C) $2\sqrt{2}-1$
(D) $\frac{1}{2\sqrt{2}}$

Q10 If $\alpha=3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta=3\cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is (are)

(A) $\cos\beta>0$
(B) $\sin\beta<0$
(C) $\cos(\alpha+\beta)>0$

(D) $\cos\alpha < 0$

Q11 Total number of solution of the equation $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \sin^{-1} x$ is/are

(A) One (B) Two
(C) Three (D) Four

Q12 Solution of $\sin^{-1}(\sin x) = \sin x$ are if x in $(0, 2\pi)$

(A) 0 (B) 1
(C) 2 (D) 3

Q13 If the area enclosed by the curves $f(x) = \cos^{-1}(\cos x)$ and $g(x) = \sin^{-1}(\cos x)$ in $x \in \left[\frac{9\pi}{4}, \frac{15\pi}{4}\right]$ is $\frac{a\pi^2}{b}$ (where a and b coprime), then find $(a - b)$

(A) -1 (B) 0
(C) 1 (D) 2

Q14 The area bounded by the curve $y = \left|\cos^{-1}(\sin x)\right| + \left|\frac{\pi}{2} - \cos^{-1}(\cos x)\right|$ and the x-axis, where $\frac{\pi}{2} \leq x \leq \pi$, is equal to

(A) π^2
(B) $\frac{\pi^2}{2}$
(C) $\frac{\pi^2}{8}$
(D) $\frac{\pi^2}{4}$

Q15 If $f(x) = \tan^{-1}\left(\frac{2^x}{1+2^{2x+1}}\right)$, then $\sum_{r=0}^{9} f(r)$ is

(A) $\tan^{-1}(1024)$
(B) $\tan^{-1}\left(\frac{1023}{1024}\right)$
(C) $\tan^{-1}\left(\frac{1023}{1025}\right)$
(D) None of these

Q16 The value $\tan\left(\sin^{-1}\left(\cos\left(\sin^{-1} x\right)\right)\right)\tan\left(\cos^{-1}\left(\sin\left(\cos^{-1} x\right)\right)\right)$, where $x \in (0, 1)$, is equal to

(A) 0 (B) 1
(C) -1 (D) 2

Q17 $4\tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right)$ is equal to

(A) π
(B) $\frac{\pi}{2}$
(C) $\frac{\pi}{3}$
(D) $\frac{\pi}{4}$

Q18 If $\cos^{-1}\left(\frac{1-a^2}{1+a^2}\right) - \cos^{-1}\left(\frac{1-b^2}{1+b^2}\right) = 2\tan^{-1} x$, then x is

(A) $\frac{a-b}{1+ab}$
(B) $\frac{b-a}{1+ab}$
(C) $\frac{a+b}{1-ab}$
(D) None of these

Q19 $\sin\left(2\tan^{-1}\left(\frac{1}{3}\right)\right) + \cos\left(\tan^{-1}(2\sqrt{2})\right)$ is equal to

(A) $\frac{14}{15}$
(B) $\frac{3}{4}$
(C) $\frac{2\sqrt{2}}{7}$
(D) $\frac{2\sqrt{2}}{15}$

Q20 The value of $\sec\left(2\cot^{-1} 2 + \cos^{-1}\frac{3}{5}\right)$ is equal to

(A) $\frac{25}{24}$
(B) $-\frac{24}{7}$
(C) $\frac{25}{7}$
(D) $-\frac{25}{7}$

Q21 The value of $\sum_{r=2}^{\infty} \tan^{-1}\left(\frac{1}{r^2-5r+7}\right)$, is

(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{2}$
(C) $\frac{3\pi}{4}$
(D) $\frac{5\pi}{4}$

Q22 If $|a| < 1, |b| < 1$ and $|x| < 1$, then the solution of $\sin^{-1}\left(\frac{2a}{1+a^2}\right) - \cos^{-1}\left(\frac{1-b^2}{1+b^2}\right) = \tan^{-1}$, is $\left(\frac{2x}{1-x^2}\right)$

(A) $\frac{a-b}{1-ab}$

(B) $\frac{1+ab}{a-b}$
(C) $\frac{ab-1}{a+b}$
(D) $\frac{a-b}{1+ab}$

Q23 The value of $\sin\left(2\tan^{-1}\frac{1}{3}\right)+\cos\left(\tan^{-1}2\sqrt{2}\right)$, is

(A) $12/13$
(B) $13/14$
(C) $14/15$
(D) none of these

Q24 If x satisfies the inequality $\left(\tan^{-1}x\right)^2+3\left(\tan^{-1}x\right)-4>0$, then the complete set of values of x is

(A) $\left(-\tan 4,\frac{\pi}{4}\right)$
(B) $(-\infty,\tan 4)\cup\left(\frac{\pi}{4},\infty\right)$
(C) $(\tan 1,\infty)$
(D) $(\tan 4,\tan 1)$

Q25 The least value of n for which $(n-2)x^2+8x+n+4>\sin^{-1}(\sin 12)+\cos^{-1}(\cos 12),\forall x\in R$, where $n\in N$, is ____________ .

(A) 4
(B) 5
(C) 6
(D) 7

Q26 $\sin^{-1}\left(\frac{4x}{x^2+4}\right)+2\tan^{-1}\left(-\frac{x}{2}\right)$ is independent of x then

(A) $x\in[-3,4]$
(B) $x\in[-2,2]$
(C) $x\in[-1,1]$
(D) $x\in[1,\infty)$

Q27 If $\cos^{-1}\frac{6x}{1+9x^2}=-\frac{\pi}{2}+2\tan^{-1}3x$, then $x\in$

(A) $\left(\frac{1}{3},\infty\right)$
(B) $(-1,\infty)$
(C) $(-\infty,0)$
(D) None of these

Q28 If $(x-1)\left(x^2+1\right)>0$, then sin $\left(\frac{1}{2}\tan^{-1}\frac{2x}{1-x^2}-\tan^{-1}x\right)$ is equal to

(A) 1
(B) $\frac{1}{\sqrt{2}}$
(C) -1
(D) None of these

Answer Key

Q1	Hence Proved	Q15	(C)
Q2	$x = \pm 3$	Q16	(B)
Q3	(C)	Q17	(D)
Q4	(D)	Q18	(A)
Q5	(D)	Q19	(A)
Q6	(C)	Q20	(D)
Q7	(A)	Q21	(C)
Q8	(D)	Q22	(D)
Q9	(B)	Q23	(C)
Q10	(B, C, D)	Q24	(C)
Q11	(B)	Q25	(B)
Q12	(B)	Q26	(B)
Q13	(C)	Q27	(A)
Q14	(D)	Q28	(C)